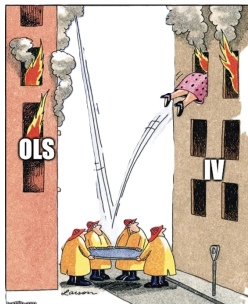


Instrumental Variable Approaches

F. Candau (UPPA-TREE)



The endogeneity problem

We want the **causal** effect β of a treatment x on an outcome y :

$$y_i = \alpha + \beta x_i + \varepsilon_i.$$

OLS recovers β only if x_i is uncorrelated with everything else in ε_i . It **fails** whenever $\text{Cov}(x_i, \varepsilon_i) \neq 0$:

- **Omitted variables** — a confounder drives both x and y .
- **Reverse causality / simultaneity** — y also affects x (supply *and* demand).
- **Measurement error** in x (attenuation).

In that case

$$\text{plim } \hat{\beta}_{OLS} = \beta + \frac{\text{Cov}(x_i, \varepsilon_i)}{\text{Var}(x_i)} \neq \beta.$$

Roadmap

Where we are going today:

- ➊ **Instrumental variables** — the two conditions, the estimator, and diagnostics.
- ➋ **Generalized Method of Moments (GMM)** — the framework behind 2SLS and behind the shares view of shift-share.
- ➌ **The shift-share (Bartik) instrument** — what it is and how it is built.
- ➍ **Identification “from the shares”** — Goldsmith-Pinkham et al. (2020).
- ➎ **Identification “from the shifts”** — Borusyak et al. (2022).
- ➏ **Inference** — Adão et al. (2019).
- ➐ **A practical guide** — which assumption fits your design?
- ➑ **Applied session** — estimating a Bartik design in Python.

Two readings underpin the whole lecture: Goldsmith-Pinkham et al. (2020) on shares, Borusyak et al. (2022) on shifts.

Section 1

Instrumental Variables

The IV solution: two conditions

An **instrument** z is a variable that moves x but is otherwise unrelated to the outcome. It must satisfy two conditions:

Relevance (testable)

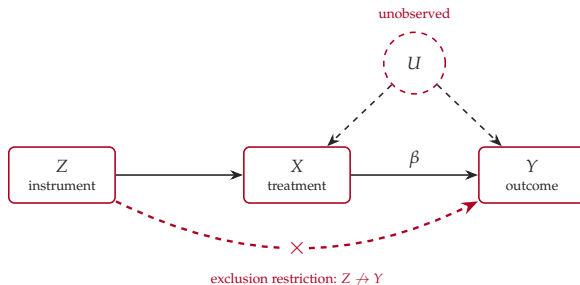
$\text{Cov}(z_i, x_i) \neq 0$ — the instrument actually shifts the treatment. Checked with the *first stage*.

Exogeneity / exclusion (not testable)

$\text{Cov}(z_i, \varepsilon_i) = 0$ — equivalently the **orthogonality (moment) condition** $\mathbb{E}[z_i \varepsilon_i] = 0$. The instrument affects y *only* through x , not through any other channel. Defended with economic argument, not a test.

The art of IV is almost entirely about **defending exogeneity**.

The logic in one picture



- **Relevance:** $Z \rightarrow X$ — the instrument moves the treatment.
- **Confounding:** U (unobserved) drives both X and Y — this is *why OLS is biased*.
- **Exclusion:** there is **no** arrow $Z \rightarrow Y$ (and none $U \rightarrow Z$) — so Z touches Y *only* through X . IV uses just the Z -driven part of X , which is clean of U .

The IV / Wald estimator

With a single instrument, take the covariance of the structural equation with z :

$$\text{Cov}(z, y) = \beta \text{Cov}(z, x) + \underbrace{\text{Cov}(z, \varepsilon)}_{=0}.$$

Solving for β gives the **IV estimator**:

$$\hat{\beta}_{IV} = \frac{\text{Cov}(z, y)}{\text{Cov}(z, x)}$$

Intuition: scale the instrument's effect on the outcome by its effect on the treatment. If z is binary, this is the **Wald estimator**, a ratio of differences in means; the name is also used for the IV ratio computed on a single contrast, such as one pair of years.

The numerator and denominator are each a **simple regression on z** : the **reduced form** (y on z) and the **first stage** (x on z). We study them next.

First stage, reduced form, and the IV ratio

The IV estimate is built from **two ordinary regressions on the instrument**:

First stage — does z move the treatment?

Regress the treatment on the instrument: $x_i = \pi z_i + \nu_i$.
The slope $\hat{\pi}$ measures **relevance**.

Reduced form — does z move the outcome?

Regress the outcome on the instrument: $y_i = \rho z_i + \eta_i$.
Under exclusion, z reaches y only via x , so $\rho = \beta \pi$.

Hence the **IV estimator** is just the **ratio** of the two:

$$\hat{\beta}_{IV} = \frac{\hat{\rho}}{\hat{\pi}} = \frac{\text{reduced form}}{\text{first stage}}.$$

If the instrument barely moves x ($\hat{\pi} \approx 0$), we divide by ~zero — the estimate explodes. This is the **weak-instrument** problem.

Two-Stage Least Squares (2SLS)

With controls w_i and possibly several instruments, the same idea is implemented in **two stages**:

- **Stage 1 (first stage):** regress x_i on the instrument(s) and controls, and keep the **fitted values**

$$x_i = \pi z_i + w_i \gamma + v_i \longrightarrow \hat{x}_i.$$

$\hat{x}_i$ is the part of x driven *only* by the instrument — purged of U .

- **Stage 2 (second stage):** regress y_i on $\hat{x}_i$ (not x_i !) and controls:

$$y_i = \beta \hat{x}_i + w_i \delta + \xi_i.$$

With a single instrument this reproduces exactly $\hat{\beta}_{IV} = \hat{\rho} / \hat{\pi}$. In matrix form, with Z the $n \times L$ matrix of the L instruments (controls partialled out) and $P_Z = Z(Z'Z)^{-1}Z'$ the projection onto them,

$$\hat{\beta}_{2SLS} = (X'P_ZX)^{-1}X'P_Zy \quad (\text{stage 1 forms } \hat{X} = P_ZX).$$

In practice, estimate 2SLS with a dedicated IV routine (Stata `ivreghdfe`; Python `feols` / `pyfixest`).

What does IV estimate? LATE and compliers

Under homogeneous effects, β is a single parameter and IV estimates it. Under heterogeneous effects β_i , the IV ratio does not identify $\mathbb{E}[\beta_i]$. It identifies the average of β_i over the subpopulation whose treatment status is shifted by the instrument, the **compliers** — the **Local Average Treatment Effect (LATE)**:

$$\hat{\beta}_{IV} \xrightarrow{p} \mathbb{E}[\beta_i \mid i \text{ complier}] \equiv \text{LATE} \quad (\text{Imbens--Angrist 1994}).$$

- The complier subpopulation is defined by the instrument and is not observed. A different instrument shifts a different subpopulation.
- Two valid instruments therefore identify two different LATEs, and a gap between $\hat{\beta}_1$ and $\hat{\beta}_2$ is not in itself evidence of invalidity.
- External validity is limited accordingly: the estimand is local to the margin the instrument shifts, not the population average effect.

Several instruments for one regressor

So far we used a single instrument z for the endogenous x . But we often have **several** valid instruments $z_1, \dots, z_L$ for the *same* x .

Excluded vs. included. The z_ℓ are the **excluded instruments**: they appear in the **first stage** (they help predict x) but are **excluded from the outcome equation**. The controls w_i , which appear in *both* equations, are the **included instruments**.

The **exclusion restriction** is exactly the claim that the excluded instruments have **no direct effect** on y — they reach y only through x .

The art of IV: defending each instrument

Example: returns to schooling

x = years of schooling, y = wages. Candidate instruments: z_1 = distance to the nearest college, z_2 = exposure to the 1959 Berthoin reform, which raised the school-leaving age from 14 to 16 for cohorts born from 1953 onwards.

For each candidate, two questions: is it **relevant** (does it move schooling? testable in the first stage), and is it **excluded** (does it reach wages only through schooling? not testable, to be argued)?

z_1 (**distance to college, Card 1995**). *Relevant?* proximity cuts the cost of attending $\Rightarrow$ more schooling (visible in the first stage). *Excluded?* it reaches wages only through schooling — credible **only after** controls for region, urbanicity and family background (college towns may differ economically).

Defending the second instrument: the Berthoin reform

z_2 (**the Berthoin reform, Grenet 2013**). *Relevant?* pupils who would have left at 14 had to stay two more years, so cohorts born from 1953 hold more schooling — a first stage visible in administrative data. *Excluded?* the date of birth is not chosen and the law applied nationally on a fixed date; the threat is that adjacent cohorts differ for other reasons (post-war expansion of secondary schools, cohort size), which is why such designs control for smooth cohort trends.

The compliers are the pupils who would have stopped at 14, so the estimate is **their** return to schooling, not the average one.

Are the instruments strong enough? The first-stage F

All excluded instruments enter the first stage together with the controls:

First stage

$$x_i = \pi_1 z_{1i} + \cdots + \pi_L z_{Li} + w_i \gamma + v_i \longrightarrow \hat{x}_i.$$

Relevance is therefore a **joint** question, and the first-stage F tests $H_0 : \pi_1 = \cdots = \pi_L = 0$.

- **Weak instruments.** Weakness is a matter of the first-stage coefficient relative to its standard error, the F , not of its size in levels. With a single instrument $F = t^2$, so an instrument **only just** significant at 5 % ($F = 4$) is still weak. 2SLS is then **biased toward OLS** and its confidence intervals **over-reject**.
- **Thresholds.** $F > 10$ as a rule of thumb (Staiger–Stock 1997); Stock–Yogo (2005) raise the threshold with the number of instruments.
- **In practice.** Use the **Kleibergen–Paap** F under heteroskedasticity or clustering, and an **Anderson–Rubin** interval when F is low, valid at any instrument strength.

Section 2

GMM

What is a “moment”?

IV is a **method of moments**. A **moment** is the expectation of a function of the data: the mean $\mathbb{E}[x]$, a covariance $\mathbb{E}[xy]$.

The method of moments, in one line

A model implies that certain moments take known values. Replace the population expectation $\mathbb{E}[\cdot]$ by its **sample average** $\frac{1}{n} \sum_i$, and solve for the parameters.

IV is exactly this. Exogeneity is a population **moment condition**:

$$\mathbb{E}[z_i \varepsilon_i(\beta)] = 0, \quad \varepsilon_i(\beta) = y_i - \beta x_i$$

(“the instrument is uncorrelated with the error”; variables are demeaned, so the intercept is dropped). We estimate β by making the **sample** version hold:

$$\frac{1}{n} \sum_i z_i (y_i - \beta x_i) = 0 \implies \hat{\beta}_{IV} = \frac{\text{Cov}(z, y)}{\text{Cov}(z, x)}.$$

One equation, one unknown β — solved **exactly**.

One instrument, or several?

One instrument (just-identified): one moment, one unknown $\Rightarrow$ exact solution (the case above).

Two instruments for the same β . In our schooling example (z_1 = distance to college, z_2 = the Berthoin reform), each is its own moment condition:

$$\mathbb{E}[z_{1i} \varepsilon_i] = 0 \quad \text{and} \quad \mathbb{E}[z_{2i} \varepsilon_i] = 0.$$

- Each *alone* gives $\hat{\beta}_1 = \frac{\text{Cov}(z_1, y)}{\text{Cov}(z_1, x)}$, $\hat{\beta}_2 = \frac{\text{Cov}(z_2, y)}{\text{Cov}(z_2, x)}$: either instrument **alone** identifies β . Together they give two equations for one unknown, and in a finite sample $\hat{\beta}_1 \neq \hat{\beta}_2$.
- The second instrument is not needed to estimate β . This surplus is **over-identification** (L instruments, K endogenous regressors: $L = K$ just-identified, $L > K$ over-identified), and it leaves $L - K$ restrictions that can be **tested**.

$\rightarrow$ **GMM** answers both: **(1)** how to *combine* them? **(2)** do they *agree* (a test of validity)?

The GMM estimator

Stack the L sample moments, with $z_i = (z_{1i}, \dots, z_{Li})'$,

$$\bar{g}(\beta) = \frac{1}{n} \sum_i z_i (y_i - \beta x_i),$$

and choose β to make them **jointly as close to zero as possible**, in the metric of a weight matrix W :

$$\hat{\beta}_{GMM} = \arg \min_{\beta} \bar{g}(\beta)' W \bar{g}(\beta)$$

- W decides **how much each moment counts**. With a single instrument there is one moment, solved exactly, and W plays no role.
- **2SLS is the case** $W = (Z'Z)^{-1}$. Another W gives another estimator of the same β : the choice affects precision, not what is identified.

Are the instruments valid?

Two valid instruments must give the same answer in the population. The **Sargan–Hansen test** asks whether $\hat{\beta}_1$ and $\hat{\beta}_2$ differ by more than sampling noise.

- H_0 : all instruments are valid. The statistic follows a χ^2 with $L - K$ degrees of freedom, one per surplus instrument.
- **Reject.** At least one instrument is invalid, or the model is misspecified. The test does not say which one.
- **Do not reject.** Reassuring, not a proof: two instruments that fail in the same direction agree with each other and pass.
- **One instrument.** No surplus, no test.

Dynamic panels: GMM with lagged instruments

The most common GMM in applied work is **dynamic panel GMM**, used when a **lagged outcome** sits on the right-hand side:

$$y_{it} = \rho y_{i,t-1} + \beta x_{it} + f_i + u_{it}.$$

- **Within breaks here** (Nickell 1981): sweeping out f_i leaves $y_{i,t-1}$ correlated with the demeaned error — biased, badly for small T .
- **Arellano–Bond (1991)**: first-difference to remove f_i , then instrument $\Delta y_{i,t-1}$ with **deeper lagged levels** $y_{i,t-2}, y_{i,t-3}, \dots$
- **Blundell–Bond (1998)** “system GMM”: also use lagged *differences* as instruments for the levels equation (needs an extra stationarity assumption).
- **We will not use these estimators.** The instruments multiply with T , the results depend on choices the reader cannot check, and the diagnostics mislead (Roodman 2009). The instruments of this chapter are observed shares and shocks, with an economic argument for their exogeneity.

Section 3

Shift-share IV

Where do good instruments come from?

The binding constraint is almost never *relevance* — it is *exclusion*.

- A valid instrument needs a credible argument for why it affects y **only** through x .
- Natural experiments (lotteries, policy discontinuities, weather...) are scarce.

Where do good instruments come from?

The binding constraint is almost never *relevance* — it is *exclusion*.

- A valid instrument needs a credible argument for why it affects y **only** through x .
- Natural experiments (lotteries, policy discontinuities, weather...) are scarce.

A recurring idea in applied work: many outcomes at the level of a *region, city, or firm* respond to **national or global shocks**, but each unit is exposed to those shocks **differently**, according to its **pre-existing composition**.

That structure — common shocks $\times$ heterogeneous exposure — is exactly the **shift-share** instrument. It is the workhorse of labor, trade, macro, public, and urban economics.

Motivating setting: local labor markets

Classic question (Bartik 1991): what is the elasticity of local **wages** y_i to local **labor demand** x_i across regions i ?

- OLS is biased: a local labor-supply shock (e.g. immigration) moves both employment and wages — $\text{Cov}(x_i, \varepsilon_i) \neq 0$.
- We need an instrument that shifts **local labor demand** for reasons unrelated to local supply.

Idea: a region's industries grow or shrink with *national* industry trends; a region tied to booming industries sees demand rise — regardless of its local supply conditions.

Notation

Throughout the shift-share part:

symbol	meaning
i	unit of analysis (region, city, firm)
$k = 1, \dots, K$	sectors / origins / “shocks”
s_{ik}	share: exposure of unit i to k (e.g. industry k ’s employment share), predetermined
g_k	shift: the common shock to k (e.g. national growth of industry k)
x_i, y_i	endogenous treatment and outcome

Shares are typically (not always) non-negative and **sum to one:** $\sum_k s_{ik} = 1$.

Construction of the instrument

The **shift-share** instrument (traditionally the **Bartik** instrument) interacts predetermined shares with common shifts:

$$z_i = \sum_{k=1}^K s_{ik} g_k$$

- *Shares* s_{ik} — measured at a **baseline** period (before the shocks).
- *Shifts* g_k — common to all units (national / global).

z_i is the **predicted** growth of unit i if each of its sectors grew at the national rate. It is then used as an instrument for the endogenous x_i .

A worked mini-example

Two sectors (manufacturing, services), two regions. Baseline shares:

region	manuf.	services
A	0.70	0.30
B	0.20	0.80

National shifts: $g_{\text{manuf}} = -10\%$ (China shock), $g_{\text{serv}} = +5\%$.

$$z_A = 0.7(-10) + 0.3(5) = -5.5\%, \quad z_B = 0.2(-10) + 0.8(5) = +3.0\%.$$

The instrument predicts region **A contracts**, region **B expands** — purely from *national trends* $\times$ *local exposure*, not from local supply shocks.

Three canonical applications

study	unit i	shares s_{ik}	shifts g_k
Bartik (1991)	region	industry employment shares	national industry growth
Card (2001)	city	origin- k migrant share, 1980	national inflow from origin k
Autor et al. (2013)	region	industry employment shares	Chinese imports (other high-income countries)

Same skeleton everywhere: **predetermined exposure** $\times$ **common shock**. The designs differ in *which component* carries the identifying assumption — the heart of this lecture.

Leave-one-out shifts

A subtle but crucial point: if the shift g_k is built from a national aggregate that **includes unit i itself**, then z_i mechanically contains i 's own outcome shock $\Rightarrow$ correlation with ε_i .

Fix: construct **leave-one-out** shifts that exclude unit i (or, as in Autor et al., measure the shock in *other countries* entirely):

$$g_k^{(-i)} = \text{growth of sector } k \text{ excluding unit } i.$$

This breaks the mechanical link and is standard practice.

The central question

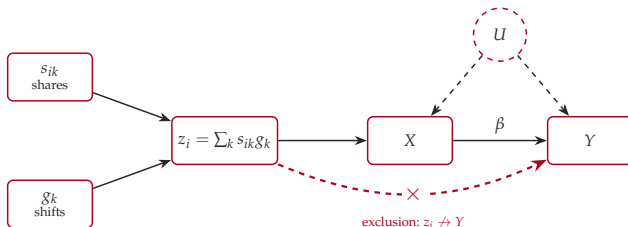
Why is z_i a valid instrument?

$z_i = \sum_k s_{ik} g_k$ has **two** ingredients. Exogeneity can be argued to come from *either* one:

- **From the shares** s_{ik} — units differ in *exposure*, and that exposure is as-good-as-random (conditional on controls). → **Goldsmith-Pinkham et al. (2020)**.
- **From the shifts** g_k — the *shocks* are as-good-as-random, even if exposure is endogenous. → **Borusyak et al. (2022)**.

These are **different research designs** with different assumptions, diagnostics, and standard errors. We take each in turn.

The two routes, in one picture



Both ingredients feed z_i ; its exogeneity (clean of U) can be earned from **either**:

- **Goldsmith-Pinkham et al. (shares):** the shares are as-good-as-random $\Rightarrow \mathbb{E}[s_{ik}\varepsilon_i] = 0$.
- **Borusyak et al. (shifts):** the shifts are as-good-as-random $\Rightarrow$ valid *even if shares are endogenous*.

Section 4

Shares view

The big idea

Goldsmith-Pinkham et al. (2020) ask: *what is a Bartik regression actually doing?*

Numerical equivalence (their Proposition 1.1)

The 2SLS estimator using the single instrument $z_i = \sum_k s_{ik}g_k$ is *numerically identical* to a GMM estimator that uses the K industry shares $\{s_{ik}\}_k$ as **separate instruments**, with a weight matrix built from the shifts.

Formally, with z the instrument, x the treatment, y the outcome (stacked over units), S the matrix of shares and W a weight matrix:

$$\hat{\beta}_{Bartik} = \frac{z'y}{z'x}, \quad \hat{\beta}_{GMM} = \frac{x'SWS'y}{x'SWS'x}.$$

With the rank-one weight matrix $W = gg'$ (the outer product of the K -vector of shifts g) and $z = Sg$ (the $n \times K$ shares matrix S times g), the two coincide: $Sgg'S' = (Sg)(Sg)' = zz'$, so $\hat{\beta}_{GMM} = \hat{\beta}_{Bartik}$.

Why this reframes identification

If Bartik = “shares as instruments”, then the **exclusion restriction lives on the shares**:

$$\mathbb{E}[s_{ik} \varepsilon_i] = 0 \quad \text{for every } k.$$

The shifts g_k are *not* instruments here — they only **combine** the K share-instruments into one. The identifying content is: *is each exposure share uncorrelated with the unobservable?*

This is the natural reading when you have **few, dominant sectors** and tell a story like “places more exposed to industry k form a clean comparison group.”

Rotemberg weights: where the estimate comes from

The Bartik estimate is a weighted average of the K single-share IV estimates, with weights $\hat{\alpha}_k$ known as the **Rotemberg weights** (Goldsmith-Pinkham et al. 2020):

$$\hat{\beta}_{Bartik} = \sum_k \hat{\alpha}_k \hat{\beta}_k, \quad \sum_k \hat{\alpha}_k = 1.$$

They show which shares produce the headline estimate.

- **Concentration is bad news.** Five industries hold more than 40 % of the weight in their application: one endogenous share among them is enough to make $\hat{\beta}$ wrong.
- **Fewer assumptions to defend.** Only those shares need an exogeneity argument; the others cannot overturn the estimate.
- **Large negative weights are a bad sign.** $\hat{\beta}$ can then take a sign that no single share supports.

Section 5

Shifts view

The big idea

Borusyak et al. (2022) start from a different intuition:

A share-weighted average of random shifts is itself as-good-as-random

As long as the shocks g_k are as-good-as-randomly assigned across many sectors, the instrument $z_i = \sum_k s_{ik} g_k$ is mean-independent of ε_i on average.

Example: skill-intensive regions may attract particular immigrants, but if subsidies are randomized across many high- and low-skill industries, exposed regions get **typical** instrument values.

The shift-share IV is a shock-level regression

Borusyak et al. (2022) show the location-level IV estimate is numerically equal to an IV regression run **at the level of the shocks**, with g_k as the instrument and each shock weighted by its share of total exposure, ω_k ($\sum_k \omega_k = 1$).

- The assumption is then stated on the **shocks**.
- The sample size that matters is the number of **shocks** K , not the number of units.
- Shocks carry unequal weight, so what counts is the effective number $K_{\text{eff}} = 1 / \sum_k \omega_k^2$. A few dominant sectors make K_{eff} small, and the design rests on a handful of shocks.

The identifying assumptions (shifts view)

Two requirements

- (1) **Quasi-random shocks (shock orthogonality):** the g_k are as-good-as randomly assigned and affect y only through x — mean-independent of the exposure-weighted unobservable.
- (2) **Many, dispersed shocks:** K_{eff} large (low exposure Herfindahl), shocks mutually uncorrelated.

Notably, **no assumption on the shares** is required — they may be endogenous. This fits “generic” industry shares (as in Autor et al. 2013) where share exogeneity would be implausible.

Controls and diagnostics (shifts view)

Make shocks conditionally as-good-as-random with the right controls:

- If shares do **not** sum to one, **control for the sum of shares** $\sum_k s_{ik}$.
- For a shock-level confounder q_k (e.g. skill intensity), control for its **shift-share aggregate** $\sum_k s_{ik} q_k$ — same shares, applied to q_k .

Diagnostics:

- **Balance of the shocks.** Regress g_k on sector characteristics — size, skill intensity, growth before the shock. A systematic relation means the shocks follow something that may also drive the outcome.
- **Concentration.** Report K_{eff} : it says on how many shocks the estimate actually rests.
- **Mechanical correlation.** When the shock is computed from data that include the unit itself, the instrument contains that unit's own outcome. Remove the unit from the computation (leave-one-out), or use a shock measured elsewhere.

Section 6

Inference

Why standard errors are wrong

A subtle problem (Adão et al. 2019): **two regions with similar exposure share vectors get similar instrument values** z_i , because z_i is the *same* linear combination of common shocks.

⇒ regression residuals are **correlated across units that look unrelated** (even in different states/clusters).

Conventional, heteroskedasticity-robust, and even region-clustered standard errors **ignore this** and badly **over-reject** — a nominal 5% test can reject 30–50% of the time.

The fix: exposure-robust standard errors

Practical guidance

Use standard errors that are robust to cross-unit correlation *induced by shared exposure*:

- **Exposure-robust** estimators of Adão et al. (2019) — valid under the exposure structure;
- equivalently, **shock-level (Borusyak et al.) standard errors** from the shift-level IV regression cluster naturally **at the level of the shocks k** ;
- both are implemented in the authors' Stata/R packages.

Rule of thumb: the unit of “independent variation” is the **shock**, not the region — cluster your thinking (and SEs) accordingly.

Section 7

Shares or shifts?

Which assumption fits your design?

Ask yourself:

- 1 **Can I defend a few shares as exogenous?** Are they *tailored* (they mediate only the shock of interest, like origin-specific migrant shares)? → **shares view**.
- 2 **Are my shares “generic”** (industry composition that proxies exposure to *many* unobserved shocks)? Then share exogeneity is implausible → lean on **exogenous shifts**.
- 3 **Do I have many, plausibly-random shocks** with low exposure HHI? → **shifts view** is available and powerful.

In practice: report **both** sets of diagnostics, and always use **exposure-robust standard errors**.

Two paths, one instrument

	Goldsmith-Pinkham et al.	Borusyak et al.
Exogenous	the shares s_{ik}	the shifts g_k
Shares may be endogenous?	no	yes
Asymptotics in	number of units	number of shocks K
Best when	few, dominant sectors; “tailored” shares	many small shocks; “generic” shares
Key diagnostic	Rotemberg weights, pre-trends	balance of g_k , K_{eff}
Inference	over-id tests; exposure-robust SE	shock-level / Adão et al. SE
Canonical	Card (2001) migration enclaves	Autor et al. (2013) China shock

Section 8

Applied

Setting and data

The applied session estimates how much an input bought by farms responds to the farm's revenue: y is the log of input spending, x the log of crop revenue, endogenous.

- **Units and indices.** i = farm, c = crop (twelve crops), t = year. The share s_{ic} is crop c 's part of the farm's crop revenue in its first two observed years; the shift is the national price of crop c in year t , $\log P_{ct}$.
- **The instrument is a panel**, $z_{it} = \sum_c s_{ic} \log P_{ct}$, used with farm and year fixed effects: within a farm it moves over time only because prices move.
- **Objects in the code.** `df`: one row per farm and year, columns `id`, `annee`, x , `y`, `prod_veg` (total crop output) and one revenue column per crop. `prix`: one row per crop and year, columns `culture`, `annee`, `log_prix`. `crops`: the list of the twelve crop columns.

```
import pandas as pd, pyfixest as pf
```

Build the two ingredients: shifts and shares

Four lines turn a price file and the panel into the shifts $\log P_{ct}$ and the shares s_{ic} .

```
lp = prix.pivot(index="annee", columns="culture",  
                 values="log_prix") # 1  
base = df.groupby("id").head(2) # 2  
tot = base.groupby("id")[crops + ["prod_veg"]].sum() # 3  
shares = tot[crops].div(tot["prod_veg"], axis=0) # 4
```

- 1 — the shifts as a table with one row per year and one column per crop: `pivot` reshapes the long price file into that grid. `lp.loc[2007, "colza"]` is a price.
- 2 — the **first two rows** of each farm, its two earliest years: the shares must be measured before the shocks they will be combined with.
- 3 — sums, over those two years, the revenue of each crop and total output.
- 4 — divides each crop's revenue by total output, row by row (`axis=0`): the share of each crop in the farm's output, one row per farm, one column per crop.

Combine them into the instrument z_{it}

Five lines attach the shares to every year of the farm and sum shares times prices.

```
pan = df.merge(shares, left_on="id", right_index=True)      # 1
pan["z"] = 0.0                                              # 2
for c in crops:                                           # 3
    pan[f"z_{c}"] = pan[c] * lp.loc[pan["annee"], c].values # 4
    pan["z"] += pan[f"z_{c}"]                             # 5
```

- 1 — joins the shares to the panel: `df` has a column `id`, `shares` has the farm as its row index, so each farm's twelve shares are copied onto all of its rows.
- 2 — starts the instrument at zero.
- 3, 4 — for each crop, the contribution $\text{share} \times \log \text{price}$ is stored in a column `z_{<crop>}`, used later for the Rotemberg weights. `pan[c]` is the farm's share of crop `c`; `lp.loc[pan["annee"], c]` picks, for each row, the price of `c` in that row's year; `.values` drops the index so that the two line up row by row.
- 5 — adds it to `z`. After the loop, z is $\sum_c s_{ic} \log P_{ct}$.

First stage, reduced form, 2SLS

Four lines estimate the three regressions of the chapter and the first-stage F .

```
cl = {"CRV1": "id"}  
fs = pf.feols("x ~ z | id + annee", pan, vcov=cl) # 1  
rf = pf.feols("y ~ z | id + annee", pan, vcov=cl) # 2  
iv = pf.feols("y ~ 1 | id + annee | x ~ z", pan, vcov=cl) # 3  
F = (fs.coef()["z"] / fs.se()["z"]) ** 2 # 4
```

- 1 — the first stage: the endogenous regressor on the instrument, farm and year fixed effects absorbed. Its coefficient is $\hat{\pi}$.
- 2 — the reduced form: the outcome on the instrument. Its coefficient is $\hat{\rho}$.
- 3 — 2SLS in one call. The formula has **three parts** separated by |: the outcome and its exogenous regressors (1 means none besides the fixed effects), the fixed effects, then endogenous $\sim$ instrument. The coefficient on x is $\hat{\rho} / \hat{\pi}$.
- 4 — with one instrument, the first-stage F is the square of its t .
- `cl` — standard errors clustered by farm; the exposure-robust errors of the inference section are not available in `pyfixest`.
- To test an elasticity of 1 rather than 0: `(iv.coef()["x"] - 1) / iv.se()["x"]`.

One shock at a time: the Wald estimator

Four lines compute the ratio of two differences on a single pair of years.

```
w = pan[pan["annee"].isin([2006, 2007])] # 1
w = w.pivot(index="id", columns="annee",
             values=["y", "x", "z"]).dropna() # 2
d = pd.DataFrame({"dy": w["y"][2007] - w["y"][2006],
                  "dx": w["x"][2007] - w["x"][2006],
                  "dz": w["z"][2007] - w["z"][2006]}) # 3
wald = pf.feols("dy ~ dz", d).coef()["dz"] \
      / pf.feols("dx ~ dz", d).coef()["dz"] # 4
```

- 1 — the two years around the shock.
- 2 — one row per farm, one column per year and variable; a farm missing a year gets a gap, dropped with `.dropna()`.
- 3 — the change of each variable between the two years, for each farm.
- 4 — reduced form over first stage on those changes: the Wald estimator, identical to 2SLS with one instrument. Its first-stage F tells whether the shock spread farms enough.

Which crops carry the estimate, and a placebo

Five lines compute the Rotemberg weights and build the placebo, next year's instrument.

```
def r(v): return pf.feols(f"{v} ~ 1|id + annee", pan).resid() # 1
xt = r("x") # 2
num = {c: r(f"z_{c}") @ xt for c in crops} # 3
alpha = {c: num[c] / sum(num.values()) for c in crops} # 4
pan["z_lead"] = pan.groupby("id")["z"].shift(-1) # 5
```

- **1** — a function that regresses a variable on the fixed effects only and returns the residuals: the variable **net of** farm and year effects. **2** applies it to x .
- **3** — for each crop, the covariance between its contribution to the instrument, $z_c = s_{ic} \log P_{ct}$, and x ; $@$ is the dot product of two vectors.
- **4** — each covariance over their sum: the Rotemberg weight α_c . They sum to one; a negative one is possible.
- **5** — z on the **next row of the same farm**, `shift(-1)` moving values up one row inside each farm: next year's prices. Added to the reduced form, a significant coefficient would mean farms react to prices before they occur.

References I

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